

Shivom Gupta

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🌐 [Sakshi Sense](#)

SUMMARY

Interdisciplinary researcher with a background in **Astrophysics (UNIGE)** and **Computational Modeling**, currently focused on the intersection of **Cognitive Neuroscience and Mantra Meditation**. I am dedicated to uncovering the latent neural representations of consciousness through the quantitative analysis of high-density EEG data. My current work investigate how **mantra-based meditation** influences neuroplasticity and physiological resilience, utilizing **EEG microstates** as a framework for understanding discrete brain state transitions. I leverage advanced skills in **Bayesian modeling**, **Gaussian Processes**, and **time-series analysis** to infer internal representations from noisy physiological signals.

SKILLS & EXPERIENCES: Python (Advanced), MATLAB, Bayesian Statistics, Gaussian Processes, Supervised Machine Learning, Time-Series Analysis, PsychoPy, MCMC, MNE, Pycrostates, Neurokit, Behavioral Experiment Design, Physiological Data Collection, Writing Pre-processing pipelines, 32-channel Neurosoft, 8 channel EEG cap, Polar device, Shimmer Device.

EDUCATION

Master in Astrophysics, UNIGE, Switzerland
(14.09.2020 - 16.06.2022)

Thesis: [Reflected Light from Hot Jupiters as Seen by TESS](#). (Please download the file for better viewing.)

Bachelors in Aeronautics, AKTU, India
(02.07.2012 - 14.06.2016)

Thesis: Co-Flow jet flow control aerofoil.

RESEARCH EXPERIENCE

Research Associate (08.09.2024 – Present)

IKS Research Centre, ISS, Delhi, New Delhi, India

- * Research Associate under the supervision of Dr. Jyotiranjana Beuria (PI), focusing on quantitative analysis of neural time-series data.
- * Data collection and analyse of multi-channel (8, 32) EEG data across meditation, mind-wandering and various research areas.
- * **Develop reproducible pre-processing and analysis pipelines.**
- * Two co-authored conference proceedings, to be presented at MBCC 2026, IIT Mandi.
- * Current project that I lead investigates how mantra-based meditation influences neuroplasticity and enhances physiological resilience in ageing. It also aims to examine the role of semantics. See a list of [projects](#).
- * Teaching a month-long course on 'Basics of Cognitive Neuroscience' (Science and Experiment) to undergrads from IIT.
- * [Sakshi Sense](#) project, contributing to experimental design, data validation, and statistical modeling of cognitive states in collaboration with the IIT Mandi team.

Python MNE Experiment Design EEG Data Collection
Time-Series Analysis Microstates PsychoPy
Bayesian Modeling Science Communication

Thesis Student (20.09.2021 - 16.06.2022)

Geneva Observatory, UNIGE, Switzerland

- * A 9-month-long [project](#), supervised by Prof. Monika Lendl and Dr. Adrien Deline.
- * Combined Gaussian Processes and a transit model (BATMAN) to detect eclipses from hot Jupiters.
- * Reported 4 new & 8 marginal detections, 12 non-detections and improved ($>3\sigma$) 3 marginal detections.

Python Gaussian Process Photometry MCMC

Research Assistant (01.08.2021 - 31.08.2021)

Geneva Observatory, UNIGE, Switzerland (Job)

- * Supervised by Prof. Daniel Schaerer.
- * Wrote a Python script to automate the LyC galaxy emitters' data visualization from different surveys.

Python Big Data Data Visualization

Project Student (22.02.2021 - 04.06.2021)

Geneva Observatory, UNIGE, Switzerland

- * A semester-long [project](#) on computing the maximum greenhouse effect for terrestrial planets.
- * Used an MH-sampler and performed computations with 30 greenhouse gases to find the best mixture.

Python Atmosphere Modeling DACE MCMC

Project Student (14.09.2020 - 18.12.2020)

Geneva Observatory, UNIGE, Switzerland

- * A semester-long [project](#) supervised by Prof. François Bouchy & Dr. Nolan Greives.
- * Worked on developing first wavelength solution for the Near-Infrared Planet Searcher (NIRPS) spectrograph.

Python Instrumentation Spectroscopy DS9



PUBLICATIONS/CONFERENCE

- * **Predicting attentional lapses during meditation:** Ultra-short-term autonomic markers for real-time biofeedback., (**Accepted for MBCC 2026**, IIT Mandi) (June 3-6, 2026) (Peer-reviewed, Oral)
- * **Contactless acoustic respiratory monitoring for Pranyama biofeedback using deep learning.**(**Accepted for MBCC 2026**, IIT Mandi) (June 3-6, 2026) (Peer-reviewed, Oral)
- * "Infrared Spectroscopy: A method for remote sensing of water and other bio-signature gases on Earth-like planets". (**Oral**), 2nd COSPAR Symposium, Brazil. (November 09-13, 2015)
- * "Detection of Marsquakes by seismic probes on-board Mars-2020 mission: An indirect way to detect life on Mars". (**Poster**), 2nd COSPAR Symposium, Brazil. (November 09-13, 2015)
- * A week-long funded "**NARIT School on Exoplanets & Astrobiology**". (April 22-26, 2019).



SCHOLARSHIP/FUNDING

- * **CHF 8800 scholarship** for Master year 2 by the Hans Wilsdorf Foundation, Geneva.
- * **Conference** (2nd COSPAR Symposium, Brazil) **travel fund of INR 160000** from the college.
- * U.P **government scholarship of INR 250000** for undergraduate tuition and living expenses.



CO-CURRICULAR/VOLUNTEER

- * **Organizing Member:** *Unriddling Inference* (Joint conference by ISS and Brhat, 2026).
- * **Volunteer:** *The Enigma of Perception* (ISS Delhi Conference, 2025).
- * Attended Astrophysics Colloquium, University of Geneva (2020-2022).
- * **Invited Speaker** (Two times): Astronomy Summer School organized by India's oldest amateur astronomy club JVP, where I spoke on exoplanets and **careers in astronomy**.
- * **Invited Speaker:** Talk on "Astronomy & Its Relation to the Environment" to INSPIRE fellow students (Grade 12) at IUCAA.
- * **Invited Speaker:** Talk on "Exoplanets and the Prospects of Finding Life" to all the students from the pre-final round of the Astronomy Olympiad at MESC, Pune.
- * **Volunteer** in, "TLC workshop on Night Sky Photometer Fabrication", **NSPF 2019**, IUCAA, Pune, India.
- * **Volunteer** in the Workshop on, "Developing Astronomy Themed Experiments (ATE) for College/Universities", **ATE 2018**, IUCAA.
- * edX Community **Teaching Assistant** (Super-Earth And Life)
- * **9 Online Courses** in Astronomy & Astrophysics (Edx &

Coursera)

- * Undergrad **Teaching Assistant** (Space Sciences)



JOB EXPERIENCE

Lodha Genius Program (15.05.2026 - 15.06.2026) [Current, 1 month]

Ashoka University, Faculty (Alien Earths: In Search of Life Beyond Our Planet)

- * Co-teaching this course with Dr. Dolon Bhattacharyya for 10 outstanding high-school and senior high-school students selected Pan India
- * Supervised and guided two undergraduate TAs across the academic and logistical aspects of the course.
- * Led a two-day educational field trip to the ARIES Observatory, enriching students' exposure to observational astronomy.

Curriculum Design

Scientific Mentorship

Leadership

Sakshi Sense (01.12.2025 - Present)

Startup's Science Lead

- * Developed a robust microphone-based breathing count algorithm for noisy physiological recordings, validated on the Breath-My dataset.
- * **Gaussian Processes Joint Modeling:** Reviewed and **fixed** the **old pipeline** for Gaussian processes joint models used in latent states study. Prototyping of the instrument is done. The improved code will be used in the Sakshi Sense project of ISS.

Young India Fellowship Program(19.05.2025 - 20.06.2025)

Ashoka University, TA (Exploring the Universe)

- * Managed a diverse class of 46 fellows, ensuring smooth coordination between faculty and students.
- * Graded assignments and assisted faculty in evaluating final presentations.
- * Organized Discussion Sessions and Office Hours with faculty and myself for deeper engagement.
- * Oversaw course logistics, communication, and technical setup to ensure seamless delivery.

Astronomy

Organization

Collaborative Coordination

Assessment

Academic Facilitation

Lodha Genius Program (17.05.2025 - 16.05.2025)

Ashoka University, Program Instructor (Alien Earths: In Search of Life Beyond Our Planet)

- * Co-taught this course with Dr. Dolon Bhattacharyya for 8 outstanding senior secondary students.
- * Supervised and guided two undergraduate TAs across the academic and logistical aspects of the course.
- * Led a two-day educational field trip to the ARIES Observatory, enriching students' exposure to observational astronomy.

Curriculum Design Scientific Mentorship Leadership

STEM Education (01.07.2022 - 30.03.2025)

AstonDesigns (Founder)

- * A social venture to promote STEM education, bridge the Art-STEM gap and empower women artists.
- * Collaborated with the European Astronomical Society (EAS) and other organizations to promote our product.

Leadership Organization Survey Fundraising

Astronomy Outreach (16.02.2018 - 28.06.2020)

Inter-University Centre for Astronomy & Astrophysics (IUCAA), Pune, India

- * Managed and mentored in public outreach and training programs for students and teachers across India.
- * Promoted projects that IUCAA is involved in, including TMT, LIGO-India, AstroSat and ADITYA-L1 missions.

CoralDraw Astrometrica Stellarium Photoshop

Astronomy Education (17.08.2016 - 12.12.2017)

Space Technology & Education Private Limited, New Delhi, India

- * Conducted astronomy workshops and national and international campaigns in various schools in Delhi.
- * Initiated NASA's SallyRide EarthKAM and 'Project Paridhi' in schools involving over 50000 students.

CoralDraw Astrometrica Stellarium



REFEREES

- * [Dr. Jyotiranjana Beuria](#), ISS Delhi.
- * [Dr. Adrien Deline](#) (Thesis Co-supervisor), University of Geneva.
- * [Prof. Daniel Schaerer](#), University of Geneva.
- * [Prof. Surhud S. More](#), Inter-University Centre for Astronomy and Astrophysics (IUCAA)